

Ullr's Secret Core Calculations Reference

Overview

This document outlines the physical formulas and theoretical basis for the calculations implemented in `core.py`.

1 Vapor Pressure and Humidity

1.1 `saturation_vapor_pressure(t_celsius)`

The saturation vapor pressure e_s (in hPa) at a given temperature T (in °C) is calculated using the Magnus formula:

$$e_s(T) = 6.112 \exp\left(\frac{17.67T}{T + 243.5}\right)$$

1.2 `get_dew_point_from_rh(temp_f, rh)`

First, the actual vapor pressure e is determined from the relative humidity (RH in %):

$$e = e_s(T) \times \frac{RH}{100}$$

The dew point temperature T_d (in °C) is derived by rearranging the Magnus formula:

$$T_d = \frac{243.5 \ln(e/6.112)}{17.67 - \ln(e/6.112)}$$

1.3 `get_rh_from_dew_point(temp_f, dew_point_f)`

The relative humidity is calculated from the ratio of actual vapor pressure (saturation vapor pressure at the dew point T_d) to the saturation vapor pressure at air temperature T :

$$RH = \left(\frac{e_s(T_d)}{e_s(T)}\right) \times 100\%$$

2 Wet Bulb Temperature

2.1 `psychrometric_equation(tw_c, t_c, e, p_hpa)` and `wet_bulb_f(...)`

The wet bulb temperature T_w is solved using the psychrometric equation:

$$e_s(T_w) - P \cdot A \cdot (1 + 0.00115T_w) \cdot (T - T_w) - e = 0$$

where P is the atmospheric pressure (hPa) and $A = 6.66 \times 10^{-4}$ is the psychrometric constant. The root-finding algorithm (`fsolve`) finds the T_w that satisfies this equation.

3 Atmospheric Pressure

3.1 `pressure_at_elevation(elevation_ft)`

Standard atmospheric pressure P (hPa) at a given elevation h (meters) is approximated by:

$$P(h) = 1013.25 \left(1 - 2.25577 \times 10^{-5} h\right)^{5.25588}$$

4 Solar Position

4.1 `estimate_solar_position(lat, lon, dt_aware)`

The solar elevation angle α and azimuth angle γ_s are calculated from the current UTC time and location.

$$\begin{aligned} b &= \frac{2\pi}{365}(N - 81) \quad (N \text{ is day of year}) \\ \delta &= 23.45^\circ \sin(b) \quad (\text{Declination}) \\ \text{EOT} &= 9.87 \sin(2b) - 7.53 \cos(b) - 1.5 \sin(b) \quad (\text{Equation of Time}) \\ \text{LST} &= \text{UTC Hour} + \frac{\text{Longitude}}{15} + \frac{\text{EOT}}{60} \quad (\text{Local Solar Time}) \\ \omega &= 15^\circ(\text{LST} - 12) \quad (\text{Hour Angle}) \end{aligned}$$

Elevation α :

$$\sin(\alpha) = \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \cos(\omega)$$

Azimuth γ_s :

$$\cos(\gamma_s) = \frac{\sin(\delta) - \sin(\alpha) \sin(\phi)}{\cos(\alpha) \cos(\phi)}$$

5 Radiative Equivalent Temperatures

5.1 `calculate_radiative_equivalent_temps(...)`

To estimate the impact of solar and thermal radiation on snow surface temperatures, we compute equivalent temperature shifts $T_{SW,eq}$ (shortwave) and $T_{LW,eq}$ (longwave) using a heat transfer coefficient $K_{rad} = 8.0$.

Shortwave Equivalent

The angle of incidence θ_i on a slope (angle β , aspect γ_a):

$$\cos(\theta_i) = \cos(\beta) \sin(\alpha) + \sin(\beta) \cos(\alpha) \cos(\gamma_s - \gamma_a)$$

Net shortwave radiation (SW_{net}) incorporating cloud cover fraction c and surface albedo α_s :

$$SW_{net} = 1361 \cdot \cos(\theta_i) \cdot (1 - 0.65c^2) \cdot (1 - \alpha_s)$$

$$T_{SW,eq} = \frac{SW_{net}}{K_{rad}}$$

Longwave Equivalent

Clear sky emissivity ϵ_{clear} depends on vapor pressure e_a :

$$\epsilon_{clear} = 0.51 + 0.066\sqrt{e_a}$$

Total atmospheric emissivity $\epsilon_{atm} = (1 - c)\epsilon_{clear} + c$. The total incoming longwave radiation considers the Sky View Factor ($SVF = \cos^2(\beta/2)$) and terrain emissivity ($\epsilon_{terrain} = 0.98$):

$$LW_{net} = \sigma [SVF \cdot \epsilon_{atm} T_{air}^4 + (1 - SVF) \cdot \epsilon_{terrain} T_{air}^4 - T_{snow}^4]$$

where $T_{snow} = 273.15$ K and $\sigma = 5.67 \times 10^{-8}$.

$$T_{LW,eq} = \frac{LW_{net}}{K_{rad}}$$

6 Effective Temperature

6.1 effective_temperature_f(...)

The final snow effective temperature combines the thermodynamic wet-bulb temperature with the radiative shifts:

$$T_{eff} = T_w + T_{SW,eq} + T_{LW,eq}$$

7 Snow Consolidation Physics

7.1 Unit Dimensionality Conversion

The integrals of effective temperature calculated in the model are natively in Fahrenheit-hours (F-hrs). However, standard hydrological and physical coefficients for snow melt and ice growth expect Degree-Days Celsius ($^{\circ}\text{C-days}$). The conversion factor is:

$$1\ ^{\circ}\text{C-day} = 1.8 \times 24 = 43.2\ \text{F-hrs}$$

Thus, an integral $I_{\text{F-hrs}}$ is converted to $I_{\text{C-days}}$ via:

$$I_{\text{C-days}} = \frac{I_{\text{F-hrs}}}{43.2}$$

7.2 Physical Density and Coefficients

Real snow density ρ is calculated using Snow Water Equivalent (SWE, mm) and physical snow depth h_0 (cm):

$$\rho = \frac{\text{SWE}/10}{h_0}$$

The dynamic percolation coefficient K_M (for melting) and freeze conductivity coefficient K_F (for refreezing) are linearly scaled by the physical density:

$$\begin{aligned} K_M &= 2.0\rho + 0.1 \\ K_F &= 10.0\rho + 0.5 \end{aligned}$$

7.3 Melt and Refreeze Depths

The physical melt depth D_{melt} (in cm) follows the Degree-Day Method:

$$D_{melt} = K_M \cdot I_{m,C\text{-days}}$$

where $I_{m,C\text{-days}}$ is the melt integral ($T_{eff} > 32^\circ\text{F}$) converted to Degree-Days Celsius.

The physical refreeze depth D_{freeze} (in cm) representing crust thickness follows a generalized Stefan Phase Change Equation:

$$D_{freeze} = K_F \sqrt{I_{f,C\text{-days}}}$$

where $I_{f,C\text{-days}}$ is the absolute freeze integral ($T_{eff} < 32^\circ\text{F}$) converted to Degree-Days Celsius.